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Lehrbuch der Algebra
Working English Translation Batch 02
General Group Theory, ğğ2–6

Translator's note

This is a working English translation prepared from the cleaned Kimi-typeset Weber source currently in hand. It covers the group-theory sections ğğ2–6 of the extracted Weber volume. Mathematical notation has been normalized lightly, but the structure and claims of the source have been preserved.

ğ2. Intersection of two groups

Cosets do not themselves have the defining properties of a group. For, in order that the composition of two elements of Q_1 should again yield an element of Q_1, one would need, for example,

$$a_1 b\, a_2 b = a_3 b,$$

and hence

$$a_1 b a_2 = a_3, \qquad b = a_1^{-1} a_3 a_2^{-1}.$$

Thus b would, contrary to the assumption, have to belong to Q. The designation *coset* is therefore to be understood only in an inexact or analogical sense.

Two divisors Q, Q' of a group P always have the element 1 in common. They may, however, have further elements in common, and these common elements form a group, which we denote by R. For if a and b belong both to Q and to Q', then by the group property of Q and Q' the same is true of the product ab; therefore R is a group. This common group is called the greatest common divisor of Q and Q', or, with a geometric turn of phrase, the *intersection* of Q and Q'.

ğ3. Normal subgroups

If Q is a divisor of P that is identical with all of its conjugate divisors, then Q is called a *normal subgroup* of P.

The group consisting only of the single element 1 is a normal subgroup of every group. We obtain another normal subgroup by taking the greatest common divisor R of all divisors conjugate to some divisor Q of P.

Indeed, it has already been proved that R, as the intersection of several divisors of P, is itself a group. Now let

$$Q,\ Q',\ Q'', \dots$$

be the divisors conjugate to Q, and let b be any element of P. Then the system of groups

$$b^{-1}Qb,\ b^{-1}Q'b,\ b^{-1}Q''b, \dots$$

is not different from the original system. If R is the intersection of the first system, then $b^{-1}Rb$ is the intersection of the second system, and consequently $R = b^{-1}Rb$; that is, R is a normal subgroup of P.

If N is a normal subgroup of a group P and b is any element of P, then

$$b^{-1}Nb = N, \qquad \text{or equivalently} \qquad Nb = bN.$$

If P has order n, N has order ν, and index μ, so that $n = \mu\nu$, then we may choose the μ elements $1, b_1, \dots, b_{\mu-1}$ in such a way that the decomposition of P into cosets is

$$P = N + Nb_1 + Nb_2 + \cdots + Nb_{\mu-1} = N + b_1N + b_2N + \cdots + b_{\mu-1}N.$$

ğ4. Composition of subsets

Let A and B be any two collections of elements of a group P (whether groups or not). By the symbolic product AB we understand the set of all elements obtained by composing each element a of A with each element b of B.

The two equations

$$AA = A, \qquad AB = BA,$$

where A is fixed and B is an arbitrary subset of P, express that A is a normal subgroup of P. For from the first relation A is itself a divisor of P, and from the second it follows for every element $b \in P$ that

$$b^{-1}Ab = A,$$

that is, A is normal in P.

If A and B are two divisors of P, then one of the two equalities

$$AB = BA, \qquad \text{or equivalently} \qquad BA = AB,$$

is the necessary and sufficient condition for the product set AB to be a group.

Because of the defining property of normal subgroups, N commutes with every b, and therefore

$$b_\nu N = Nb_\nu,$$

so that

$$Nb_\nu \cdot Nb_\mu = NN \cdot b_\nu b_\mu.$$

Since $b_\nu b_\mu$ lies in P, the set $Nb_\nu b_\mu$ coincides with one of the cosets occurring in the decomposition above; and because $NN = N$, it follows that if we write $Nb_\nu b_\mu = M$, then

$$NM = M.$$

Thus property (1) in the definition of a group has been verified for the composition of cosets. That the associative law also holds was already noted. It remains only to verify the analogue of cancellation.

ğ5. Multiple isomorphism

These quotient groups are related to one another in the same way as isomorphic groups in the ordinary sense, except that to each element A there may correspond not just one element a, but several. This gives occasion to extend the notion of isomorphism by the following definition.

A group P with elements $a, b, c, \dots$ is said to be *multiply isomorphic* to a group Q with elements $A, B, C, \dots$ if the two groups can be put into correspondence in such a way that to each of the elements $A, B, C, \dots$ there correspond one or more of the elements $a, b, c, \dots$, while every element of Q corresponds to one and only one element of P, and such that whenever a corresponds to A and b corresponds to B, then ab corresponds to AB.

In this way one proves first that to each of the elements of the quotient group there corresponds an entire coset of the original group. Following H"older, one writes such a quotient group in the form P/N. The order of the complementary group is equal to the index of the group N.

We mention in conclusion a theorem about normal subgroups. If N is a divisor of P, and N' is a divisor of N which is at the same time a normal subgroup of P, then N' is also a normal subgroup of N. This is immediate, because for every element $b \in P$ one has

$$b^{-1}N'b = N',$$

and every element of N is also an element of P.

The converse, however, is not valid in general: if N is a normal subgroup of P and N' is a normal subgroup of N, it need not follow that N' is a normal subgroup of P.

ğ6. Composition series and the theorem of C. Jordan

A group P is called *simple* if it has no normal subgroup other than itself and the identity; it is called *composite* if other normal subgroups are present.

A normal subgroup which is not contained properly in any larger proper normal subgroup of P is called a *maximal normal subgroup* of P. If P_1 is a maximal normal subgroup of P, P_2 is a maximal normal subgroup of P_1, and so on, then one obtains a chain

$$P,\ P_1,\ P_2,\ \ldots,\ P_\nu = 1,$$

in which each subgroup is a maximal normal subgroup of the preceding one. Such a chain is called a *composition series*; the corresponding indices are its indices of composition.

The important theorem expressing the uniqueness of the composition factors was established by C. Jordan.

Theorem (Jordan). *However a composition series of a group P may be chosen, the sequence of its indices is always the same up to order.*

The proof rests on the quotient groups introduced above. Suppose Q and Q_1 are normal subgroups of P, and that in addition Q_1 is a divisor (hence, by the previous section, a normal subgroup) of Q. Then the quotient group Q/Q_1 is a normal subgroup of P/Q_1, and the index of Q/Q_1 in P/Q_1 is equal to the index of Q in P.

This reduction is the crucial device in the proof that the composition factors are independent of the chosen composition series, apart from their order.